

RENANCE

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# FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

AMS 102 — Past Questions & Materials  
**by Renance**

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Course: AMS 102  
Institution: Federal University of Petroleum Resources, Effurun  
Papers in this pack: 1  
Levels: 100

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This study pack was compiled and restructured by Renance for AMS 102 students at Federal University of Petroleum Resources, Effurun. It brings together 1 past paper(s)/material(s) into one clean, readable document so you can revise without chasing scattered PDFs. Practise these same questions interactively on Renance with instant answers, AI explanations, and full CBT mock exams — over 235,000+ questions across JAMB, Post-UTME, WAEC, NECO and university courses.

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## Paper 1 of 1 — AMS 102

Level 100 • First Semester • PastQuestion

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$5 Y= 2x43 \text{ oe Mw ral ave } 2 \text{ ae}$

$: 2 \S \text{ex:} + 3) \{a\} 2 + 2x2$

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$6. y= (5 + 2x2)$

$\text{rs: (a) } 2x \text{ } 20 \text{ (b) } \_\_\_ 12x \text{ \#} 20 \_\_\_$

$=x^\circ + 35x + 7) (\sqrt{2.3} \text{ } 5x77 \text{ (} x? + 2x? + 5x + 7$

$\text{te) } 8x + 25 \text{ (d) } 8x - 25$

$(x' \text{ } 2x? + 5x + 7 \text{ (e+ aut } 4 \text{ Se } 7y¥$

$7 \text{ jee Se (a) } \_\_\_ 15x \_\_\_ \text{ (b) } 30x \text{ (c) } \_\_\_ 45x \_\_\_ \text{ (d) } 60x \_\_\_$

$(6\text{---} \text{ xy } (6\text{--}77 \text{ (} 6\text{--}x^*)" \text{ (} 6\text{--}x^*y \text{ (} 6\text{--}79"$

$6. \text{ a c (a) } 2x \text{ --- } 2\text{ef}X^*2K+3) \text{ (yy g(X*-2N43) (6) } 2 \text{ --- Dye IX-2KFB)}$

$(x? \text{ --- } 2x43)$

$+, \text{ ee } |$

$(\text{d) } 2\text{ç}x\text{--}1)\text{e}^* \text{ } 8"$

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$(\text{d) } (4x? \text{ --- } 6x^*)\log.{}^{\text{™}} \sim \text{--}^* )$

$\text{ei ivative of the function: } y = 3x^*\text{---} 2x^* + x? \text{ } 1 |$

$\text{f ate gis } 2x \text{ (6) } 1808 \sim 48x \text{ (c) } 60x \sim 24x? + 2 \text{ (d) GON? --- } 249? + 2x$

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in fis xy'ds (a)  $Sx'' + C$  (b)  $(3 W4C)$  (c) r

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20, fos lx? +H) etx (a)  $30 C$  iby  $x' + x' + h + Clepxt + x''$  '

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$2 ix + 1$  )  $lx^{\otimes} + 2x$  )  $1 + C$  (ope ext exe

$(x? + Ju + 2)$  ds f 4  $\zeta$  2 Z

22 The probability that a pirl d | Ltification wrongly is f und the probability that he does her division correct! { What is the probability th t 'uate  $5 \ll 4$ ) correctly and evalunt rong) (a) 0.65 (b) 08

; Two numbers are chosen at random from the three nurmt 1 the probahility that the sum of the two is Hot odd a) (b) qd

24 \ pair of dice are throw: hat is the probability of vetting « tote' re of

> probebitt 0.1, for a National

= , ER selected for a club team 180-1," ted for at least

' lbe tr on comet ors os rte prbebt o S BORE

' one of the teams? = (a) 0.125 (b) 0.250 (c) 0,175

| Age in years [10 a

Number of Pupils G 27 7 class. What is the

The table above shows the number of pupils in each age group eB probability that a pupil chosen at random is at least 11 years old?

(0) "Ap (>) Pho (©) 0

30. A survey was carried out on 100 families each with 6 children. distribution of the number of boys per family.

| No. of Boys 16 fo |

| Frequency 2 ae =e ists |

What is the probability that a family sedomly iy selected ed will hac at least 4 sits? wi Veo (Bb) v0 (0) \*/teo

The table below shows the

31. How long will it take to pay off a debt of N880 if N25 is paid the first month, N27 the second month and N29 the third month, etc?

(a) 12 months b) 15 months  $\zeta$ ) 18 months d) 20 months

\$2. If the first three terms of an arithmetic progression are  $x + |$ ,  $3x - 18$  and  $2x - 1$ , what is the tenth term? (a) 58 b) 68 c) 78 d) 88

33, The  $9''$  and  $22''$  terms of an arithmetic progression are 29 and 55 respectively. Find the sum of its 60 terms.(a) 3240 b) 4320 ©) 2340 Ai) 4230

34, Which term of the sequence 5, 14, 23, ..., is 239?

(a)  $25^*$  b)  $27''$  ©)  $29''$  d)  $30''$ ,

35, Compute the sum of the geometric progression 2, -4, 8, -16,... .. to the 8 term

(a) -170 b) 170 6) -190 d) 190

36. If the first term of an AP is -3 and the 8<sup>th</sup> term is 11, find the common difference.

(a) 2 (b) -1 (c) | (d) 2

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42>. The toliowing ives a geomnetr® ;

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: ig) Om) b) 1002 ey it! aby

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: 44. Compute the sum of the GP Lens! ( )4K2 58 i) 4) 72

45. Find the mun of M4, 2% Wh ' ») ot ae

M6. Find th bor wf the nequcr™ ® sh) ay et by kt op Ot 4)

47. How muany conncoutev HERE rcycuittsittge vith 100 rrrsast tre tankeerts fae thetr to he equal  
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tk. Solve the inequality  $48 = oad : b U opx a) K$

49. Totu and his be dher Obi plan to Uy f for their mother during ber binthday, Tol se

pry N400,00 ernee Mua cia. i the gh \ fat more thats N300,00, what the bar i

Ponalble amount paid by Tol!

fay' < N1700,00 Ts 41600.00 )1 < N1500.08 i) N1200.01

1. Solve the Aequatity2s ~| (a) byaed dyn)

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Find the range of value of  $f$  which =

\$2. Solve the inequality 'J'

} ; y \

53. Interest computed on 1 basis of 408 (days in of slender you all)

(a) Ordinary interest (b) Annual simple interest (c) Simple interest 1 | t interest

ded quarterly on 4 BY you, better rate sud yea will yield

in Jed round

d) Equal

Higher

ccnain for 4 yore?

value of annuity now we 200

) N2,991.813 00 per ETE on AY NDOT

SH Wiens the rate interest on 89,000 (9) for 5 years OST nye 00

da) NIKO GO ») Ni 800.49 enias 0.00 -

\$9, What is the simple interest on N0 000 for 8

» the 700.00 \*

fa) N290.00 b) N270.00 a page ay N200 .

ey

@). Find the exact interest on N00 00 for 99 days at HO PEE and

5

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for N15R.00 by N168.00 d)

eyny 78.00